

PH 101

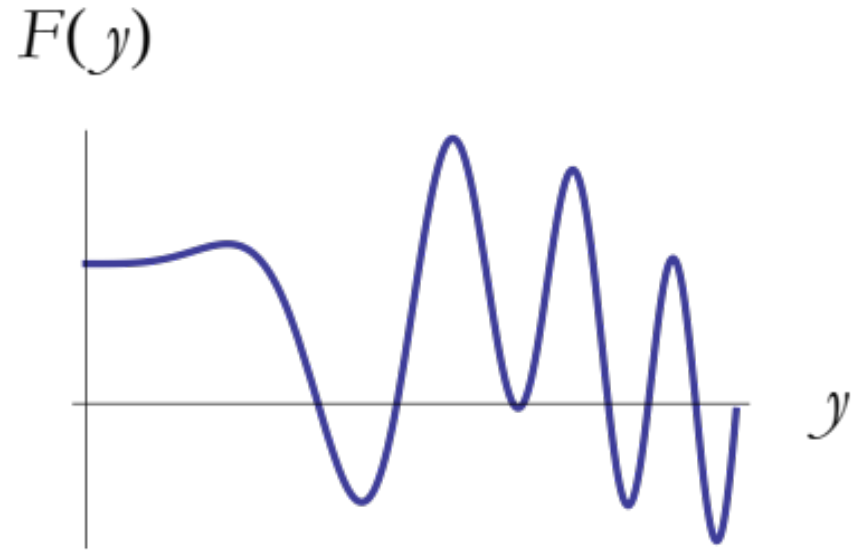
Lecture 06

OUTLINE

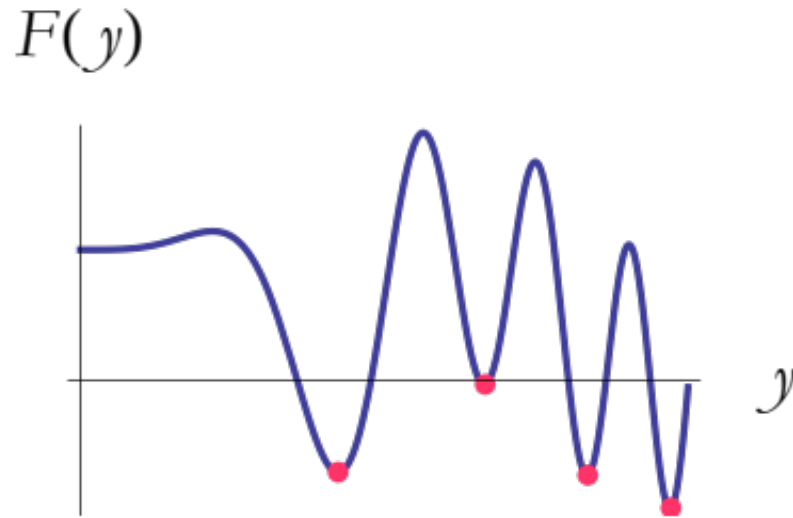
- Stationary Points and Minimizing Functions
- Calculus of Variations
- Euler's equation-I

Stationary Points and Minimizing Functions

Let's look at a function of y that we call, say **F**



Notice that there are places on the curve where a shift in y in either direction produces only an upward shift in F . These points are called *local minima*.



Local minima -> (look at the red dots)

For each local minimum, when you go in either direction along y , you begin to rise above the dot in $F(y)$. Each dot is at the bottom of a little depression. The global minimum is the lowest possible place on the curve.

One condition for a local minimum is that the derivative of the function with respect to the independent variable at that point is zero. This is a necessary condition, but not a sufficient condition. This condition defines any **stationary point**:

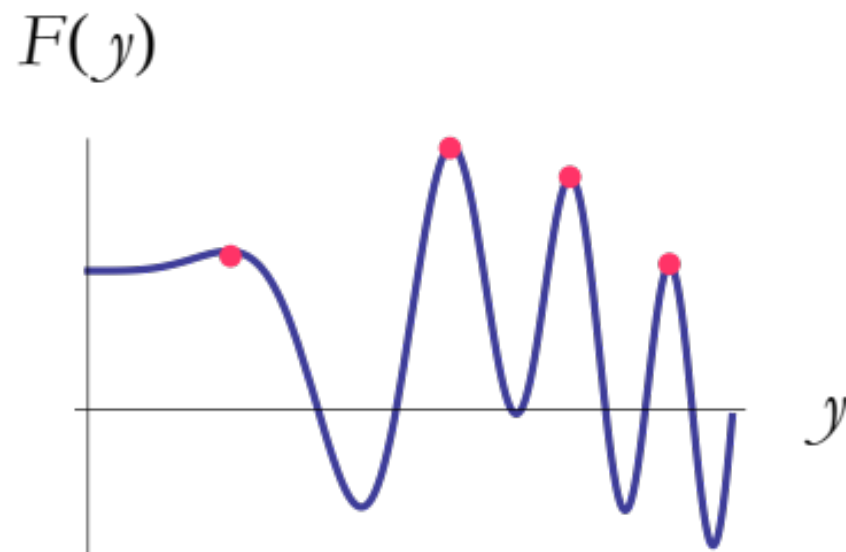
$$\frac{d}{dy}F(y) = 0$$

The second condition tests to see what the character of the stationary point is by examining its second derivative. If the second derivative is larger than 0, then all points nearby will be above the stationary point, and we have a **local minimum**:

$$\frac{d^2}{dy^2}F(y) > 0$$

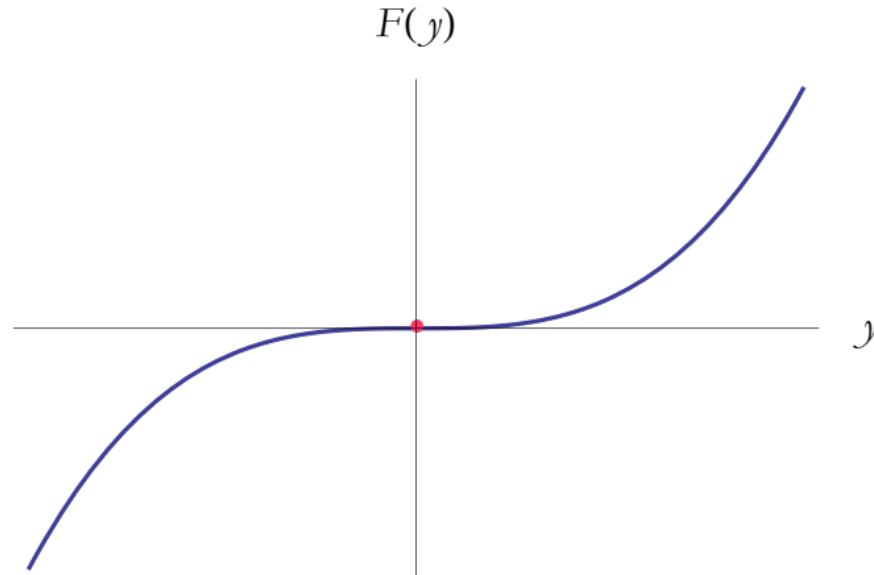
If the second derivative is less than 0, then all points nearby will be below the stationary point, and we have a **local maximum**:

$$\frac{d^2}{dy^2} F(y) < 0$$



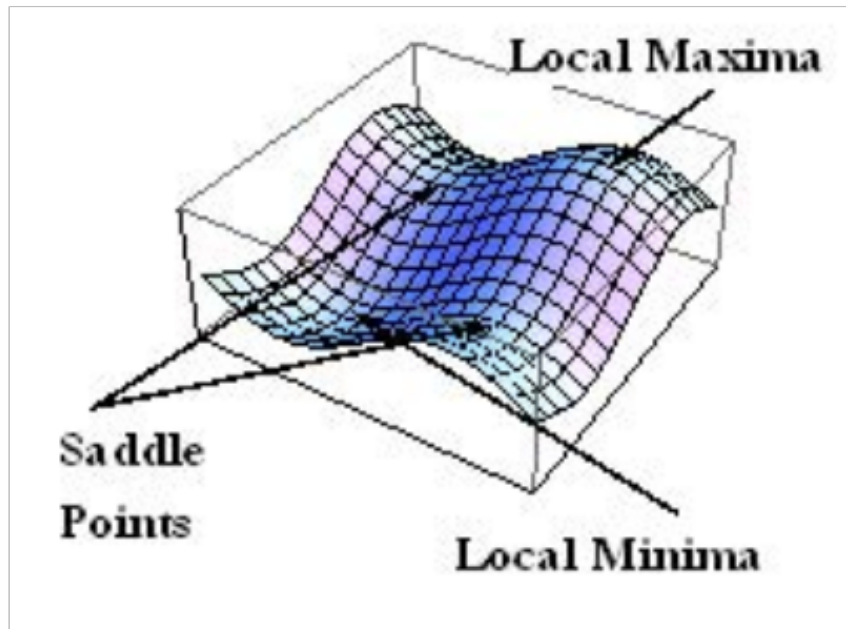
If the second derivative is equal to 0, then the derivative changes from positive to negative at the stationary point, which we call a **point of inflection**:

$$\frac{d^2}{dy^2} F(y) = 0$$

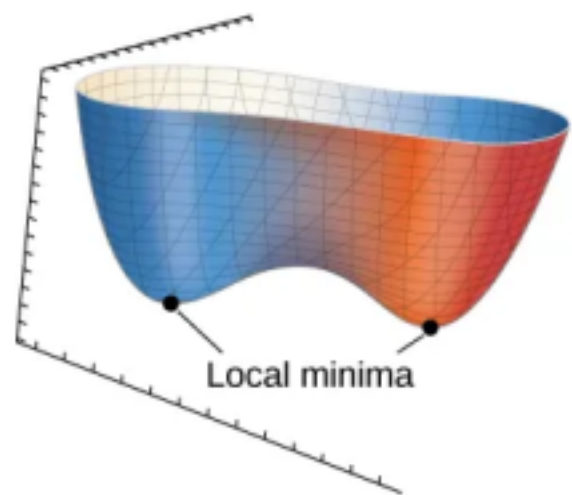


Stationary Points in Higher Dimensions

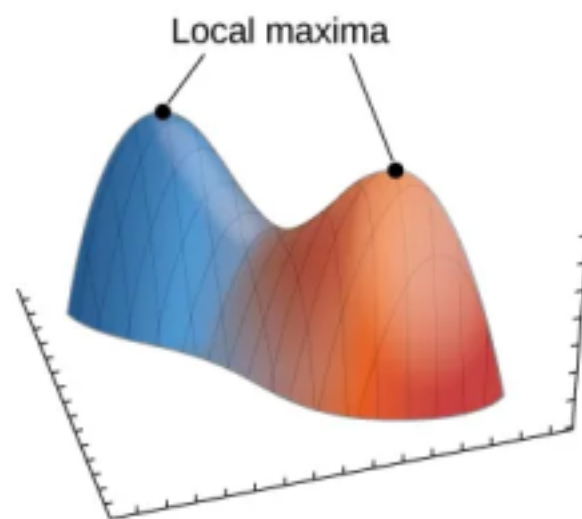
Local maxima, local minima, and other stationary points can happen for functions of more than one variable. Imagine a hilly terrain. The altitude is a function that depends on the two coordinates—let's say latitude and longitude. Call it $A(x,y)$. The tops of hills and the bottoms of valleys are local maxima and minima of $A(x,y)$. But they are not the only places where the terrain is locally horizontal. Saddle points occur between two hills.



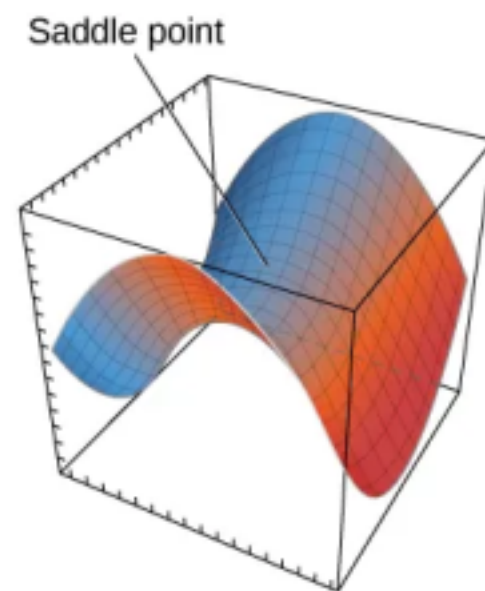
The very tops of hills are places where no matter which way you move, you soon go down. Valley bottoms are the opposite; all directions lead up. But both are places where the ground is level. There are other places where the ground is level. Between two hills you can find places called saddles. Saddle points are level, but along one axis the altitude quickly increases in either direction. Along another perpendicular direction the altitude decreases. All of these are called stationary points.



(a)

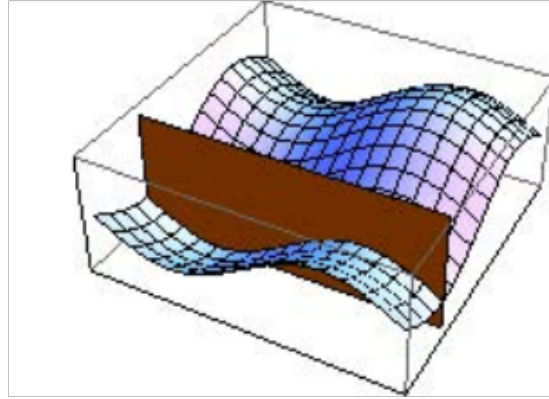


(b)



(c)

Let's take a slice along the x axis through our space so that the slice passes through a local minimum of A ,



(a slice along the x-axis)

At the minimum, the derivative of A with respect to x vanishes, we write this:

$$\frac{\partial A}{\partial x} = 0$$

On the other hand, the slice could have been oriented along the y axis, and we would then conclude that

$$\frac{\partial A}{\partial y} = 0$$

To have a minimum, or for that matter to have any stationary point, both derivatives must vanish. If there were more directions of space in which A could vary, then the condition for a stationary point is given by:

$$\frac{\partial A}{\partial x_i} = 0 \quad \text{for all } x_i \quad (1)$$

There is a shorthand for summarizing these equations. Recall that the change in a function when the point x is varied a little bit is given by

$$\delta A = \sum_i \frac{\partial A}{\partial x_i} \delta x_i$$

The set of Equations (1) are equivalent to the condition that:

$$\delta A = 0$$

for any small variation of x .

Calculus of Variations

Understanding via examples

The calculus of variations involves finding the minimum or maximum of a quantity that is expressible as an integral.

Example 1: The Shortest Path between Two Points

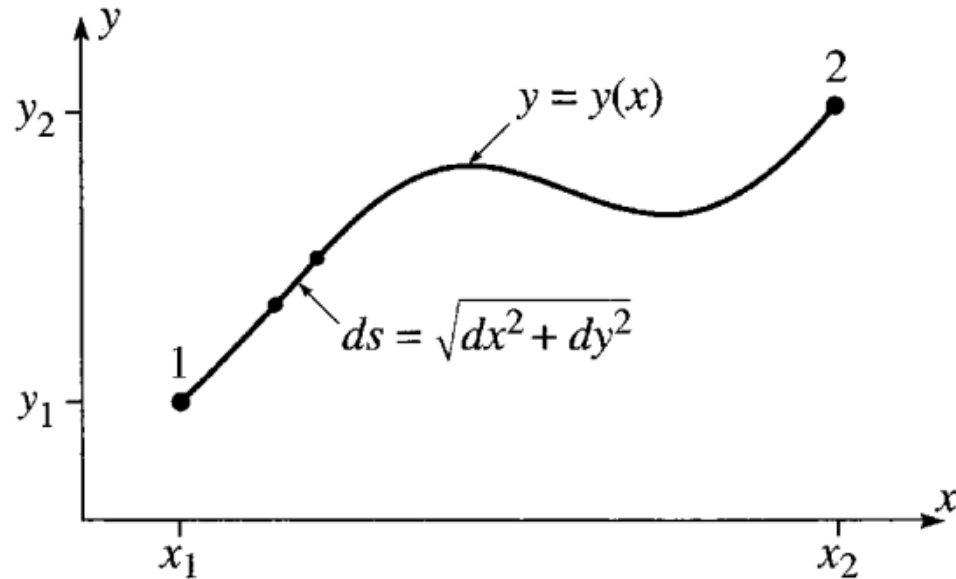
Given two points in a plane, what is the shortest path between them? While you certainly know the answer — **a straight line** — you probably have not seen a proof, unless you have studied the calculus of variations.

The length of the short segment is

$$ds = \sqrt{dx^2 + dy^2}$$

and the **total length L** of the path is

$$L = \int_1^2 ds$$



A path joining the two points 1 and 2.

$$dy = \frac{dy}{dx} dx = y'(x) dx$$

$$\begin{aligned} ds &= \sqrt{dx^2 + dy^2} = \sqrt{dx^2 + y'(x)^2 dx^2} \\ &= \sqrt{1 + y'(x)^2} dx \end{aligned}$$

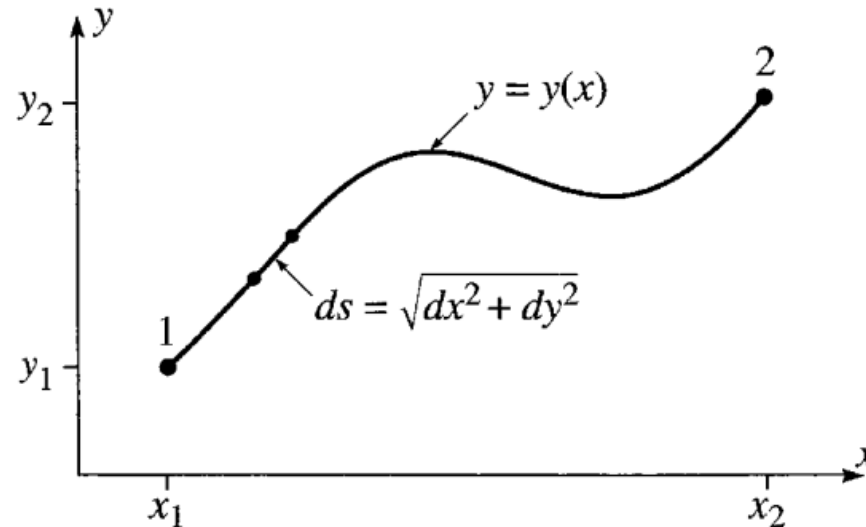
$$L = \int_1^2 ds = \int_1^2 \sqrt{1 + y'(x)^2} dx \quad (1)$$

- This equation puts our problem in mathematical form: The unknown is the function $y = y(x)$ that defines the path between points 1 and 2.
- **The problem is to find the function $y(x)$ for which the integral (1) is a minimum.**

Example 2: Fermat's Principle

A similar problem is to find the path that light will follow between two points. If the refractive index of the medium is constant, then the path is, of course, a straight line, but if the refractive index varies, or if we interpose a mirror or lens, the path is not so obvious. The French mathematician Fermat (1601-1665) discovered that the required path is the path for which the time of travel of the light is minimum.

The time for light to travel a short distance **ds** is **ds/v** where v denotes the speed of light in the medium, $v = c/n$ where n is the refractive index.



Fermat's principle says that the correct path between points 1 and 2 is the path for which the time

$$(\text{time of travel}) = \int_1^2 dt = \int_1^2 \frac{ds}{v} = \frac{1}{c} \int_1^2 n ds$$

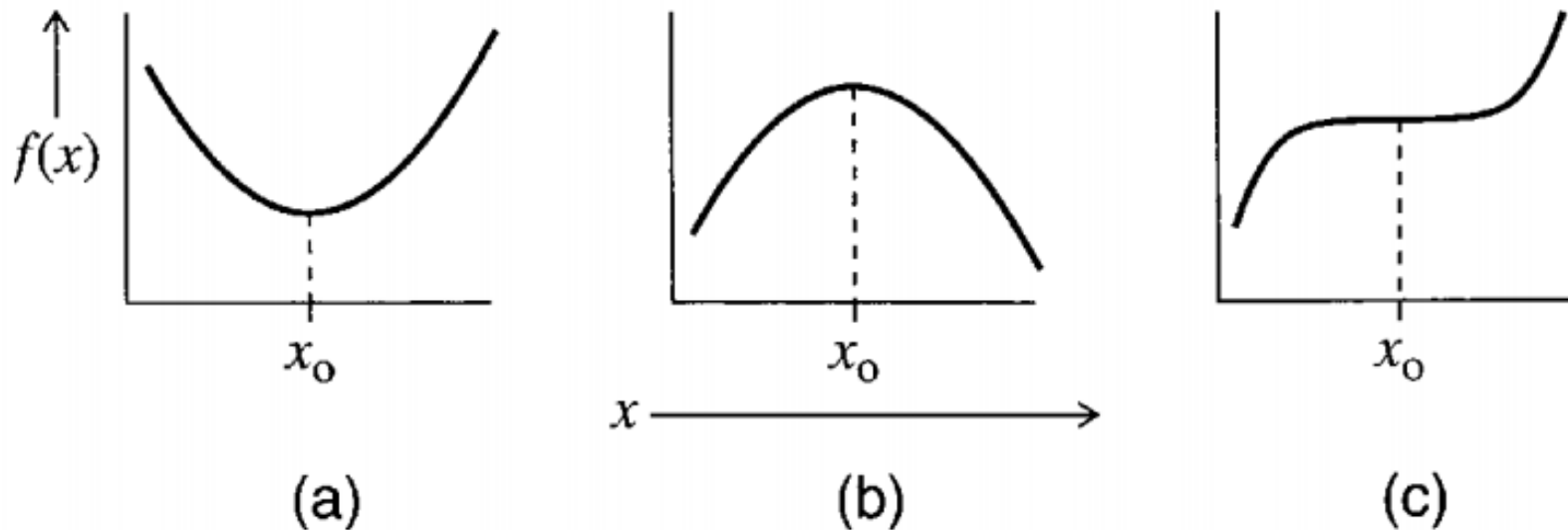
In general, the refractive index can vary, $n = n(x, y)$, and our problem is to find the path $y(x)$ for which the **integral**

$$\int_1^2 n(x, y) ds = \int_{x_1}^{x_2} n(x, y) \sqrt{1 + y'(x)^2} dx \quad (2)$$

is *Minimum*

The integral that has to be minimized in connection with Fermat's principle is very similar to **the integral (1) giving the length L of a path**; it is just a little more complicated, since the factor $n(x, y)$ introduces an extra dependence on x and y . Similar integrals arise in many other problems. Sometimes we want the path for which an integral is a maximum, and sometimes we are interested in both maxima and minima.

Think again about the problem of finding maxima and minima of functions in elementary calculus



If $\frac{df}{dx} = 0$ at x_0 , there are three possibilities: (a) If the second derivative is positive, then $f(x)$ has a minimum at x_0 . (b) If the second derivative is negative, then $f(x)$ has a maximum. (c) If the second derivative is zero, then there may be a minimum, a maximum, or neither (as shown)

The situation for the problems we are going to deal with in calculus of variations is very similar.

$$L = \int_1^2 ds = \int_1^2 \sqrt{1 + y'(x)^2} dx \quad (1)$$

$$\int_1^2 n(x, y) ds = \int_{x_1}^{x_2} n(x, y) \sqrt{1 + y'(x)^2} dx \quad (2)$$

The **method** we shall describe **now** actually finds the path that makes an integral like (1) or (2) **stationary**, in the sense that an infinitesimal variation of the path from its correct course doesn't change the value of the integral concerned.

Since our concern is how infinitesimal variations of a path change an integral, the subject is called the **calculus of variations**.

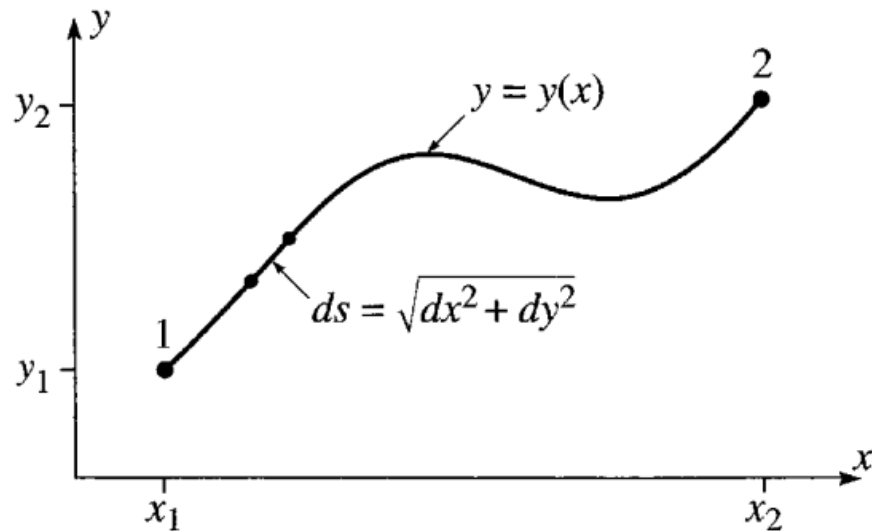
Basically, we have an integral of the form

$$L = \int_1^2 ds = \int_1^2 \sqrt{1 + y'(x)^2} dx \quad (1)$$

$$S = \int_{x_1}^{x_2} f[y(x), y'(x), x] dx \quad (3)$$

$$\int_1^2 n(x, y) ds = \int_{x_1}^{x_2} n(x, y) \sqrt{1 + y'(x)^2} dx \quad (2)$$

where $y(x)$ is an as-yet unknown curve joining two points (x_1, y_1) and (x_2, y_2) as in the Figure



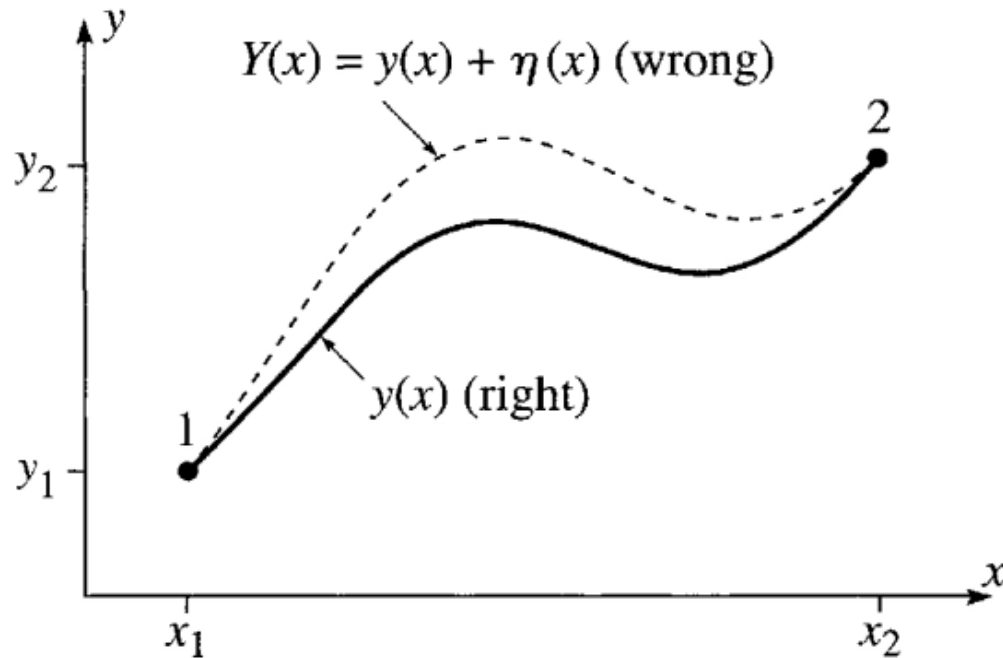
$$y(x_1) = y_1 \quad (4)$$

$$y(x_2) = y_2$$

Among all the possible curves satisfying (4) (that is, joining the points 1 and 2), we have to find the one that makes the integral **S** a **minimum** (or maximum or at least stationary).

$$S = \int_{x_1}^{x_2} f[y(x), y'(x), x] dx \quad (3)$$

To be definite, **let us suppose that we wish to find a minimum**. Notice that the function **f** in (3) is a function of three variables $f = f(y, y', x)$, but because the integral follows the path $y = y(x)$ the **integrand $f[y(x), y'(x), x]$ is actually a function of just the one variable x** .



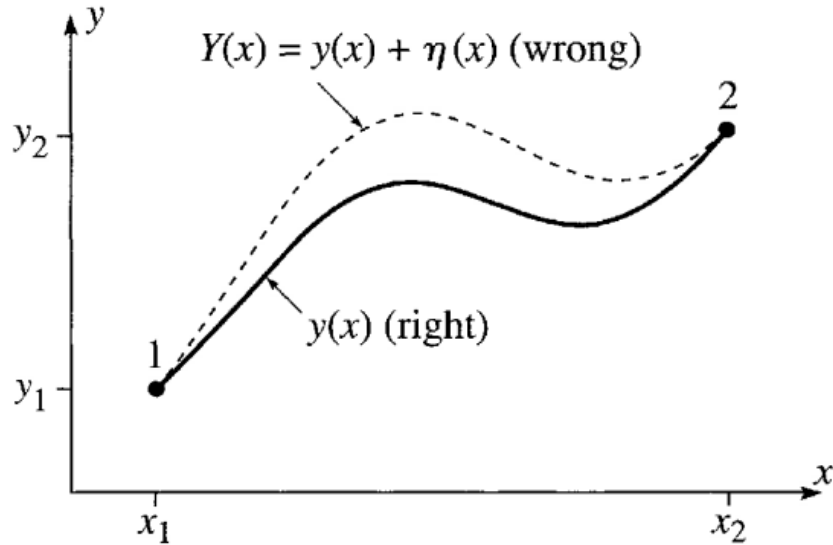
The path $y = y(x)$ between points 1 and 2 is the "right" path, the one for which the integral S of (3) is a minimum. Any other path $Y(x)$ is "wrong," in that it gives a larger value for S .

It is convenient to write the "wrong" curve $Y(x)$ as

$$Y(x) = y(x) + \eta(x) \quad (5)$$

$\eta(x)$ is just the difference between the wrong $Y(x)$ and the right $y(x)$.

Since $Y(x)$ must pass through the endpoints 1 and 2, $\eta(x)$ must satisfy



$$\eta(x_1) = \eta(x_2) = 0. \quad (6)$$

The integral S taken along the wrong curve $Y(x)$ must be larger than that along the right curve $y(x)$, no matter how close the former is to the latter. To express this requirement, we introduce a parameter α and redefine $Y(x)$ to be

$$Y(x) = y(x) + \alpha \eta(x) \quad (7)$$

The integral S taken along the curve $Y(x)$ now depends on the parameter α , so we shall call it $S(\alpha)$. The right curve $y(x)$ is obtained from (7) by setting $\alpha = 0$. Thus the requirement that S is minimum for the right curve $y(x)$ implies that $S(\alpha)$ is a minimum at $\alpha = 0$.

We have converted our problem to the traditional problem from elementary calculus of making sure that an ordinary function [namely $S(\alpha)$] has a minimum at a specified point ($\alpha = 0$).

To ensure this, we must just check that the derivative $\frac{dS}{d\alpha}$ is zero when $\alpha = 0$.

If we write out the integral $S(\alpha)$ in detail, it looks like this:

$$\begin{aligned} S(\alpha) &= \int_{x_1}^{x_2} f(Y, Y', x) dx \\ &= \int_{x_1}^{x_2} f(y + \alpha\eta, y' + \alpha\eta', x) dx \end{aligned} \tag{8}$$

To differentiate (8) with respect to α , we note that α appears in the integrand f , so we need to evaluate $\frac{\partial f}{\partial \alpha}$. Since α appears in two of the arguments of f , this gives two terms, namely (using the chain rule):

$$\frac{\partial f(y + \alpha\eta, y' + \alpha\eta', x)}{\partial \alpha} = \eta \frac{\partial f}{\partial y} + \eta' \frac{\partial f}{\partial y'}$$

and for $dS/d\alpha$ (which has to be zero):

$$\frac{dS}{d\alpha} = \int_{x_1}^{x_2} \frac{\partial f}{\partial \alpha} dx = \int_{x_1}^{x_2} \left(\eta \frac{\partial f}{\partial y} + \eta' \frac{\partial f}{\partial y'} \right) dx = 0. \quad (9)$$

This condition must be true for any $\eta(x)$ satisfying (6); that is, for any choice of the "wrong" path $Y(x) = y(x) + \alpha \eta(x)$.

Evaluation of $\frac{\partial f}{\partial \alpha}$: a more clearer explanation!

$$\begin{aligned} f &= f(\gamma, \gamma', x) \\ &= f(\gamma + \alpha \eta(x), \gamma' + \alpha \eta'(x), x) \end{aligned}$$

α is an independent parameter, it does not depend on x , γ and γ' .

$$\frac{\partial f}{\partial \alpha} = \frac{\partial f}{\partial \gamma} \frac{\partial \gamma}{\partial \alpha} + \frac{\partial f}{\partial \gamma'} \frac{\partial \gamma'}{\partial \alpha} + \frac{\partial f}{\partial x} \underbrace{\frac{\partial x}{\partial \alpha}}_{=0}$$

$$\begin{aligned}
 \text{Now, } \frac{\partial \gamma}{\partial \alpha} &= \frac{\partial (\gamma + \alpha n(x))}{\partial \alpha} = \frac{\partial \gamma}{\partial \alpha} + \frac{\partial \alpha}{\partial \alpha} n(x) \\
 &= 0 + n(x) \\
 &= n
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial \gamma'}{\partial \alpha} &= \frac{\partial (\gamma' + \alpha n')}{\partial \alpha} = \frac{\partial \gamma'}{\partial \alpha} + \frac{\partial \alpha}{\partial \alpha} n' \\
 &= 0 + n' = n'
 \end{aligned}$$

$$\text{Thus, } \frac{\partial f}{\partial \alpha} = \eta \frac{\partial f}{\partial \gamma} + \eta' \frac{\partial f}{\partial \gamma'}$$

Again:

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} \frac{\partial y}{\partial x} = \frac{\partial f}{\partial y} (y + \alpha x)$$
$$= \frac{\partial f}{\partial y}$$

Similarly,

$$\frac{\partial f}{\partial x'} = \frac{\partial f}{\partial y'} \frac{\partial y'}{\partial x'} = \frac{\partial f}{\partial y'} (y' + \alpha' x')$$
$$= \frac{\partial f}{\partial y'}$$

Hence,

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} \alpha + \frac{\partial f}{\partial y'} \alpha'$$

$$\frac{dS}{d\alpha} = \int_{x_1}^{x_2} \frac{\partial f}{\partial \alpha} dx = \int_{x_1}^{x_2} \left(\eta \frac{\partial f}{\partial y} + \eta' \frac{\partial f}{\partial y'} \right) dx = 0. \quad (9)$$

NOTE: $[\eta' = \frac{d\eta}{dx}]$

Let us rewrite the **second term** on the right using integration by parts

$$\int_{x_1}^{x_2} \eta'(x) \frac{\partial f}{\partial y'} dx = \left[\eta(x) \frac{\partial f}{\partial y'} \right]_{x_1}^{x_2} - \int_{x_1}^{x_2} \eta(x) \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) dx \quad (10)$$

Because, $\eta(x_1) = \eta(x_2) = 0$, the first term on the right (the "endpoint term") is zero.

Thus:

$$\int_{x_1}^{x_2} \eta'(x) \frac{\partial f}{\partial y'} dx = - \int_{x_1}^{x_2} \eta(x) \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) dx. \quad (11)$$

Substituting this identity into (9), we find that

$$\int_{x_1}^{x_2} \eta(x) \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} \right) dx = 0 \quad (12)$$

This condition must be satisfied for any choice of the function $\eta(x)$

Clearly, the **factor in large parentheses** must be zero:

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} = 0$$

for all x (in the relevant interval $x_1 < x < x_2$). This is the so-called **Euler-Lagrange equation** (named for the Swiss mathematician Leonhard Euler, 1707-1783, and the Italian-French physicist and mathematician Joseph Lagrange, 1736-1813) , which lets us find the path for which the integral S is stationary

More about Eq. (12)

$$\int_{x_1}^{x_2} \eta(x) \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} \right) dx = 0 \quad (12)$$

Equation (12) has the form $\int \eta(x)g(x)dx = 0$. We would certainly not claim that this condition alone implies that $g(x) = 0$ for all x . However, (12) holds for any choice of the function $\eta(x)$, and if $\int \eta(x)g(x)dx = 0$ for any $\eta(x)$, then we can conclude that $g(x) = 0$ for all x . To prove this, we must assume that all functions concerned are continuous, but, as physicists, we would take for granted that this is the case! Now, to prove the assertion, let us assume the contrary, that $g(x)$ is nonzero in some interval between x_1 and x_2 . Then choose a function $\eta(x)$ that has the same sign as $g(x)$ (that is, η is positive where g is positive and η is negative where g is negative). Then the integrand is continuous, satisfies $\eta(x)g(x) > 0$, and is nonzero at least in some interval. Under these conditions $\int \eta(x)g(x)dx$ cannot be zero. This [contradiction](#) implies that $g(x)$ is zero for all x .

